

Job Change: A Naturally Occurring Field Experiment¹

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A naturally occurring field experiment of job change was conducted over a one-year period in three applied research and development organizations. The experimental group (n = 42) were those employees who had been promoted or assigned to a different job. Employees who had maintained the same job made up the nonequivalent control group (n = 166). Standardized change score analysis showed the job change group to have greater increases in performance, innovativeness, and job satisfaction, as well as a greater reduction in role ambiguity when compared to the nonequivalent control group. In addition, the job change group reported greater increases in the job characteristics of variety and task autonomy, as well as in the role requirements of integration and boundary-spanning activities. The importance of the results and the role of field experimentation are discussed.

INTRODUCTION

Job design has become a major focus of organizational researchers and practitioners because the topic is central to the important concerns of organizational effectiveness and development as well as those of individual performance and satisfaction. A basic proposition of job design theorists and researchers has been that a significant job change will affect an array of variables related to the performance and satisfaction of individuals and to organizational effectiveness. The purpose of the present research was to investigate the effects of significant job changes by way of a naturally oc-

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curing field experiment. A job change in this research consisted of either a promotion or an assignment to a different job classification and duties without an increase in organizational level.

Various job characteristics have been related to such job outcomes as performance and satisfaction by the often-reviewed theory of Hackman and Oldham (1975) (Aldag & Brief, 1979; Griffin, 1979; Nemiroff & Ford, 1976; Pierce & Dunham 1976; Schwab & Cummings, 1976; Steers & Mowday, 1977). A basic hypothesis generated from this theory is that greater amounts of the job characteristics (i.e., variety, autonomy, task identity, feedback, dealing with others, and friendship) will lead to higher performance, satisfaction, and work motivation. In general, the cited literature reviews have found moderate support for the Hackman and Oldham theory. The majority of the reviewed field research, however, has used cross-sectional surveys. Consequently, researchers have called for more carefully conducted research that: (1) examines the effects of job changes in an organizational context; (2) does not depend solely on self-report measures; and (3) uses a longitudinal research design.

Although there is a relative lack of field experimentation reported in the job design literature, there have been some laboratory experiments with student samples that have been recently reported. O'Reilly and Caldwell (1979) manipulated tasks by enriching them and measured job characteristics with the Job Characteristic Inventory (JCI) and the Job Diagnostic Survey (JDS)—the two most popular measures of job characteristics that are based on the Hackman and Oldham (1975) theory. The results indicated that these perceptual instruments are as much affected by informational cues as they are by objective job change. Stone (1979) found the perceptions of job characteristics by subjects who had completed a standardized task were significantly influenced by "field independence"; that is, a person's disposition to being influenced by such factors as the social and physical context in perceiving job characteristics. Finally, White and Mitchell (1979) had subjects work on a simulated clerical task and found that social cues given by co-workers as well as the physical properties of the task affected perceptions of job enrichment and job ambiguity as measured by a JDS-type instrument.

Survey research by Brief and Aldag (1978), O'Reilly, Parlette, and Bloom (1980) and Dubin, Porter, Stone, and Champoux (1974) also has raised concerns about the substantive validity of perceptual job characteristic measures and the ability of the JCI or JDS to measure the effects of actual job changes. Brief and Aldag (1978) examined the JCI with a sample of 155 nurses and did not find 6 distinct factors nor convergence between superior and incumbent ratings of the same job. Dubin et al. (1974) studied 16 jobs from a telephone company and found differential perceptions of the same job across incumbents, peers in other jobs, and supervisors. A modified version of the JDS was used in this research. O'Reilly et al. (1980)

surveyed 76 nurses and found frames of reference (i.e., age, income, tenure in unit, education, and father's education) and job satisfaction to affect JDS scores.

The concerns about the JCI and the JDS are important because these instruments are widely used by researchers as well as by managers and consultants for the diagnosis and evaluation of job design and enrichment programs. Managers concerned about instituting such programs generally want to be sure that the instrument used will accurately measure the effects of job changes on employees in an organizational setting.

NATURALLY OCCURRING FIELD EXPERIMENTS

The call for true experiments in field settings is continuously made by organizational researchers, yet very few true field experiments are conducted in organizations. Rather, the preponderance of the literature from field settings is from one-time correlational studies. The majority of studies that are longitudinal in nature, moreover, usually lack an experimental treatment and must infer causality in a passive way.

A major barrier to true field experiments, as noted by Lawler (1977), is the requirement of random assignment of subjects to treatment and control groups. Organizational realities preclude this random assignment in most cases since management cannot reasonably promote, pay, lead, motivate, or otherwise treat employees on a random basis while still attempting to operate an effective organization. Ethical considerations also argue against random assignment to treatments for employees who earn their livelihood from their jobs. Other problems pointed out by Lawler include requirements for the control group not to be affected by the treatment in any way, limiting the experiment to the variables studied, and controlling for all other variables in the organization.

Lawler (1977) argues, however, that quasiexperimental designs using nonequivalent control groups (Cook & Campbell, 1979) can be effectively used in organizational settings to increase our understanding of organizational behavior in a more powerful way than the typical one-time correlational study. These designs are called "adaptive experiments" by Lawler and can be either naturally occurring or part of an induced change in the organization. Because adaptive experiments have greater external validity, Lawler also argues that they sometimes can be superior to true experiments in an organization because they do a good job of controlling for such threats to internal validity as history, maturation, testing, and regression. Kenny (1975) takes the position that nonequivalent control group designs are often the only way to evaluate real-world treatments and for social scientists to contribute to the public good.

Lawler (1977) sets out the requirements for a sound adaptive experiment to be: (a) a broad range of standardized measures, (b) before and after data on all subjects, (c) a control or comparison group, and (d) a researcher who is independent of the treatment success. Ettlie (1977) and Kenny (1975) present similar requirements for this type of research.

The present research lent itself well to a quasiexperimental, nonequivalent control group design. The researchers were conducting a longitudinal study of technological innovation in research and development organizations; hence, broad-based, standardized measures were obtained from all subjects on a range of important variables without unacceptable testing effects. The treatment was real and important in the judgment of management, yet the subjects were not otherwise affected since the treatment was naturally occurring. A nonequivalent control group was readily available, and the researchers were not involved in the treatment itself. Realistically, this research could not have been conducted as a true experiment since none of the executives in the organizations involved would allow promotions and important job assignments to have been made on a random basis. The high external validity of this type of research, however, suggests that it can yield useful information.

The present research was conducted because of the need for field experimentation in the investigation of job design and the concern about the ability of perceptual job characteristic instruments to measure the effects of actual job changes. The theme of the present research was captured in the form of two basic research questions:

Question 1. Do job changes affect such variables as employee performance and job satisfaction?

Question 2. Is a perceptual job characteristic instrument capable of measuring the effects of actual job changes?

METHOD

Data were collected on the same measures twice, one year apart. Methods used included self-report instruments, management ratings, and peer nominations for each of the two data collection points. Since data were obtained on job title, duties, subunit affiliation, and level in the organization, the researchers were able to determine who had been either promoted or assigned to a different job classification and duties during the year. Those subjects who had experienced a job change ($n = 42$) therefore constituted the experimental group while those subjects who maintained the same job ($n = 166$) made up the nonequivalent control group. The rate and types of job changes were typical of those for the past five years in the organizations studied.

Subjects

Professional employees from three research and development organizations, all located in the southwestern United States, were subjects for the present research. One organization was the research and development laboratory for a large manufacturer of capital equipment for the petroleum industry. The other two organizations were departments of a large governmental research and development complex. One department developed computer software systems for use in aerospace projects, while the other department developed power and propulsion systems for aerospace vehicles. These three organizations represented a range of applied research and development efforts.

The sample of 256 professional employees for the first data collection point consisted of 91% of the target population in the three organizations. One year later, 208 professional employees (81% of time 1) participated again, and these employees constituted the sample for the present research. Given normal turnover and absenteeism, this response rate is acceptable for a longitudinal study. Eighty-nine percent of the sample was male. The respondents had an average age of 40 years, and had been in their present position for an average of six years. Almost all of the subjects held a bachelor's degree, and most of them had some graduate education.

Measures

Job-related *performance* was measured by management ratings. Management rank-ordered the professional employees within their work unit on each of the following five performance criteria: quality of performance, quantity of performance, ability to get along with other employees, dependability, and total performance. The score on each criterion was the employee's rank-order divided by the number of employees in the work unit. A varimax factor analysis of the five performance criteria indicated that only one clear factor existed; this factor was called overall performance. The reliability (coefficient alpha) of this overall performance measure (sum of the five criteria) in the present sample was .93.

Innovativeness was also chosen as an important indicator of performance because the basic mission of a research and development organization is to generate innovations. Based on Zaltman, Duncan, and Holbek (1973), an innovation was defined as "an important product or process, or idea leading to a new or improved product, that is new to your organization." Illustrative examples were given for each organization to ensure that a common concept of innovation was understood. Innovativeness was measured by peer nominations (sociometric choice). A recent review of peer assessment methods by Kane and Lawler (1978) concluded that peer nomination is an accurate and reliable method to

measure important organizational variables. Each subject was asked to nominate up to four co-workers who had contributed most to important innovations in their respective organization. The number of nominations received by each individual became the innovativeness score for that person (Lindzey & Byrne 1968; Stogdill & Shartle, 1955).

Satisfaction was measured by the Job Descriptive Index (JDI), which was developed and validated by Smith, Kendall, and Hulin (1969). The five sub-scales of the JDI (satisfaction with the work itself, supervision, pay, promotional opportunities, and co-workers) were summed to obtain an overall measure of job satisfaction. As suggested by Smith et al. (1969), a summation of the JDI scales can be useful in a longitudinal project such as the present research. The JDI has been shown to have acceptable reliability and validity across a number of different samples (Golembiewski & Yeager, 1978; Smith et al., 1969). The reliability (coefficient alpha) of the satisfaction measure in the present sample was .87.

Job characteristics were measured by the Job Characteristic Inventory (JCI) of Sims, Szilagyi, and Keller (1976). The JCI is a widely used, 35-item instrument designed to measure perceived job characteristics included in the Hackman and Oldham (1975) theory. The JCI has been shown to have acceptable reliability and validity across diverse samples (Pierce & Dunham, 1978; Sims et al., 1976). A varimax factor analysis for the present sample revealed a four-factor solution similar to the results of Brief and Aldag (1978). The four JCI scales, with reliabilities (coefficient alpha) for the present sample given in parentheses, were as follows: *contact with others* (.84), *variety* (.86), *task autonomy* (.81), and *feedback* (.85).

The *role requirements* instrument, developed and validated by Miles (1976) specifically for research and development professionals, is a 28-item instrument which measures the extent to which various activities are part of an employee's job. This instrument was included in the present research because it makes an excellent complement to the JCI by providing measures of actual job activities (as recommended by Brief and Aldag, 1978). A varimax factor analysis of the Miles instrument for the present sample resulted in a solution that was very similar to the three-factor solution reported by Miles. (Only one item did not load on the same factor as that reported by Miles.) The three role requirements scales and their reliabilities (coefficient alpha) for the present sample were as follows: *personnel supervision activities* (.88), *integration and boundary-spanning activities* (.94), and *scientific research activities* (.70). A factor analysis of the variables in the JCI and role requirements instrument indicated that no additional interpretive value would result from a combination of these variables.

The Rizzo, House, and Lirtzman (1970) instrument of 14 items was used to measure *role conflict* and *role ambiguity*. These role variables have

been shown to be important in prior research for understanding the effects of job changes (House & Rizzo, 1972; Miles, 1976). A varimax factor analysis confirmed the two-factor solution. Reliabilities (coefficient alpha) for the current sample were .78 for role conflict and .84 for role ambiguity.

Procedure and Data Analysis

The research instruments were completed twice (one year apart) in groups of approximately 30 subjects each at the respective organizations. Management ratings and peer nominations were also collected at these times. The time lag of one year was chosen to allow sufficient time for changes in innovation to occur. Preliminary analysis showed no major differences across the three organizations on the bases of response rate, age, sex composition or education; moreover, the correlation matrices among the variables for the three organizations showed similar patterns. Based on this preliminary analysis the data were consolidated and analyses were performed for the total sample.

The standardized change score analysis recommended by Kenny (1975) for the nonequivalent control group design of this research was used to assess treatment effects. To conduct this analysis the treatment and control groups were dummy coded and then correlated with the pretest and post-test scores of a variable. The null hypothesis was tested by the equality of the two treatment-effect correlations via a test credited to Hotelling and described by Dunn and Clark (1971). A significant difference between the two treatment-effect correlations in the predicted direction means that a significant standardized change score has occurred.

While a detailed discussion is beyond the scope of the present paper, Kenny (1975) argues that a standardized change score analysis, under the proper conditions, can yield interpretable results from treatment effects with a nonequivalent control group design. Kenny's position is in contrast to that of Cronbach and Furby (1970) who stated that treatment effects could not be compared from the nonequivalent control group design because there is no way to correct for the pre-existing differences between the treatment and control groups. Kenny argues that treatment effects can be compared and interpreted through standardized change score analysis, but that there usually will be some "equivocalities" in the analysis. All inference is "risky business" whether it be from true experiments or quasiexperiments. "We cannot expect certainty from data, only fallible information" (Kenny, 1975, p. 359).

Kenny's (1975) position is reasonable and is supported by Lawler (1977) who notes that a true experiment with random assignment of subjects to treatment and control groups is precluded in most organizational

situations. If the field of organizational behavior is to use and learn from field experiments researchers will have to expect some of the equivocalities mentioned by Kenny. Of course field experiments are of value since theorists, researchers, and practitioners can gain the substantial external validity that is often lacking in true laboratory experiments.

RESULTS

The data in Table I report the raw means and the standardized change score analysis recommended by Kenny (1975). The raw means are reported only as information for the reader since they were not directly used in the tests for significant changes. Significant changes were tested by the differ-

Table I. Means and Standardized Change Score Analysis for Job Change Group and Nonequivalent Control Group

Variable	Group ^a	Mean		Difference between treatment-effect correlations
		Pre	Post	
Innovativeness	Job change	2.19	2.67	$p < .04$
	Control	3.83	3.54	
Performance	Job change	228.41	259.12	$p < .01$
	Control	272.84	271.29	
Job satisfaction	Job change	140.66	145.39	$p < .05$
	Control	149.99	151.44	
Role conflict	Job change	29.11	28.14	N.S.
	Control	27.88	27.18	
Role ambiguity	Job change	21.71	20.02	$p < .05$
	Control	20.22	20.08	
Variety	Job change	24.42	26.54	$p < .05$
	Control	26.95	26.94	
Task autonomy	Job change	35.33	39.65	$p < .02$
	Control	37.01	37.07	
Contact with Others	Job change	33.45	33.88	N.S.
	Control	33.71	34.18	
Feedback	Job change	14.61	14.57	N.S.
	Control	14.62	14.66	
Personnel supervision activities	Job change	17.73	17.86	N.S.
	Control	22.56	22.49	
Integration & boundary-spanning activities	Job change	28.97	31.33	$p < .04$
	Control	33.20	33.71	
Scientific research	Job change	8.26	8.07	N.S.
	Control	8.31	8.05	

^a $n = 42$ for job change group and $n = 166$ for nonequivalent control group.

ences between the treatment-effect correlations (not reported since the magnitudes of the correlations do not have meaning) as explained in the prior section.

Table I reports that the job change (treatment) group had significantly greater increases in performance, innovativeness, and job satisfaction, as well as a significantly greater reduction in role ambiguity when compared to the nonequivalent control group. In addition, the job change group showed significantly greater increases in the job characteristics of variety and task autonomy, as well as in the role requirements of integration and boundary-spanning activities.

The job change group also was split into those subjects with promotions ($n = 10$) and those with nonpromotional job changes ($n = 32$). These two subgroups were then compared to the nonequivalent control group via the standardized change score analysis. Tables II and III reveal

Table II. Means and Standardized Change Score Analysis for Promotion Subgroup and Nonequivalent Control Group

Variable	Group ^a	Mean		Difference between treatment-effect correlations
		Pre	Post	
Innovativeness	Promotion	2.40	3.50	$p < .01$
	Control	3.80	3.54	
Performance	Promotion	277.80	295.60	$p < .03$
	Control	272.84	271.29	
Job satisfaction	Promotion	145.70	149.90	$p < .05$
	Control	149.99	151.44	
Role conflict	Promotion	30.00	30.10	N.S.
	Control	27.88	27.18	
Role ambiguity	Promotion	24.50	21.80	$p < .04$
	Control	20.22	20.08	
Variety	Promotion	24.70	26.90	$p < .05$
	Control	26.95	26.94	
Task autonomy	Promotion	35.30	39.00	$p < .04$
	Control	37.01	37.07	
Contact with others	Promotion	34.30	35.00	N.S.
	Control	33.71	34.18	
Feedback	Promotion	14.50	13.70	N.S.
	Control	14.62	14.66	
Personnel supervision activities	Promotion	16.00	19.00	$p < .01$
	Control	22.56	22.49	
Integration & boundary spanning activities	Promotion	30.60	36.60	$p < .02$
	Control	33.20	33.71	
Scientific research activities	Promotion	9.10	9.20	N.S.
	Control	8.31	8.05	

^a $n = 10$ for promotion subgroup and $n = 166$ for nonequivalent control group.

Table III. Means and Standardized Change Score Analysis for Nonpromotion Subgroup and Nonequivalent Control Group

Variable	Group ^a	Mean		Difference between treatment-effect correlations
		Pre	Post	
Innovativeness	Nonpromotion	2.13	2.40	$p < .05$
	Control	3.83	3.54	
Performance	Nonpromotion	212.96	247.71	$p < .01$
	Control	272.84	271.29	
Job satisfaction	Nonpromotion	139.09	143.98	$p < .05$
	Control	149.99	151.44	
Role conflict	Nonpromotion	28.84	27.53	N.S.
	Control	27.88	27.18	
Role ambiguity	Nonpromotion	20.84	19.46	$p < .05$
	Control	20.22	20.08	
Variety	Nonpromotion	24.34	26.42	$p < .05$
	Control	26.95	26.94	
Task autonomy	Nonpromotion	35.34	39.85	$p < .02$
	Control	37.01	37.07	
Contact with others	Nonpromotion	33.18	33.53	N.S.
	Control	33.71	34.18	
Feedback	Nonpromotion	14.65	14.84	N.S.
	Control	14.62	14.66	
Personnel supervision activities	Nonpromotion	18.28	17.50	N.S.
	Control	22.56	22.49	
Integration & boundary spanning activities	Nonpromotion	28.46	29.68	$p < .05$
	Control	33.20	33.71	
Scientific research activities	Nonpromotion	8.00	7.72	N.S.
	Control	8.31	8.05	

^a $n = 32$ for nonpromotion subgroup and $n = 166$ for nonequivalent control group.

that the promotion subgroup's increases reached higher levels of significance for innovativeness and integration and boundary-spanning activities when compared to the significance levels of the nonpromotion subgroup. In addition, the promotion subgroup had a significantly greater increase in personnel supervision activities when compared to the nonequivalent control group, while the nonpromotion subgroup did not have a significant difference on this variable. Performance, satisfaction, role ambiguity, variety, and task autonomy had about the same levels of significance in both subgroups.

The first research question asked if job changes affect such important variables as performance and job satisfaction. The results of this field experiment give an affirmative answer to this question since job changes

preceded increases in performance, innovativeness, and satisfaction, and a decrease in role ambiguity for the research and development employees studied.

The second research question asked if a perceptual job characteristic instrument is capable of measuring the effects of actual job changes in an organization. The results showed that the job change group had significantly greater increases in the variety and task autonomy scales of the JCI, but no changes in the contact with others and feedback scales. While inferences from this field experiment are more difficult to draw when compared to a laboratory experiment with a standardized task manipulation, the increases in variety and task autonomy do seem logical. These results suggest that the JCI can measure significant amounts of true variance along with the usual method variance of a perceptual instrument.

The role requirements scale of Miles (1976) measures specific job activities. Here, the job change group (especially the promotion subgroup) showed a significantly greater increase in integration and boundary-spanning activities, and the promotion subgroup had a significantly greater increase in personnel supervision activities. These increases also make sense and indicate that the role requirements instrument also measures the effects from actual changes in important job components. Based on these results the answer to the second research question concerning whether a perceptual job characteristic instrument is capable of measuring the effects of actual job changes might best be stated as a probable yes.

DISCUSSION

The results of this field experiment indicate that a job change, with or without a promotion, can be beneficial in a number of ways. The new job can be more enriching and stimulating with greater degrees of variety, task autonomy, and integration and boundary-spanning activities. The reduction in role ambiguity may have resulted from a broader perspective of the organization and a greater understanding of its goals due to the job change. The overall result of a job change seems to be a better performing, more innovative, and more satisfied member of the organization.

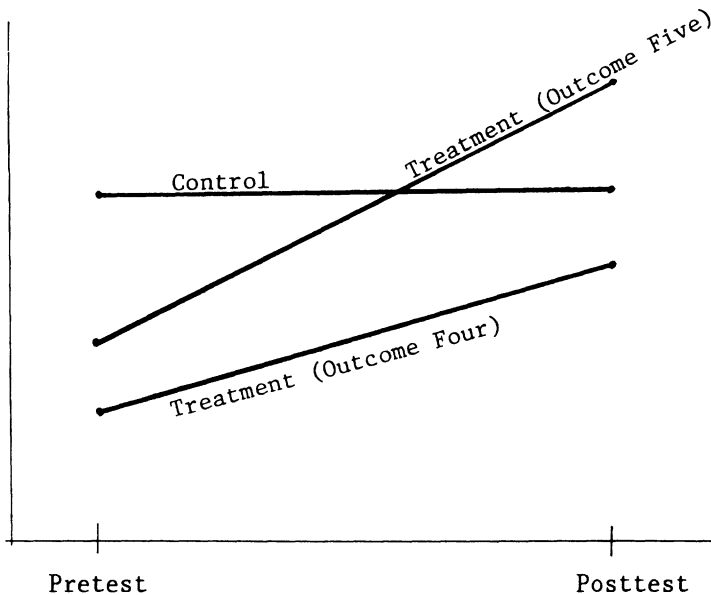
While some of these job-change results may have been due to a "novelty effect," the variables affected were sufficiently broad and important (e.g., performance, innovativeness, satisfaction) to indicate that substantial job changes occurred. In addition, the average time between the job change and the second data point was six months. This amount of time would suggest that more than just a novelty effect was at work.

The results also indicate that perceptual measures of job characteristics, such as the JCI and the activities-based role requirements instrument

(Miles, 1976), can measure the effects of actual job changes in an organization. These results are encouraging: they suggest that significant amounts of true variance were measured even though these instruments have the usual method variance associated with almost all paper-and-pencil measures of organizational variables. Continued work on these instruments is still needed to provide more stable factors and to further minimize error variance; however, such work appears warranted since these instruments serve important purposes in the diagnosis and evaluation of job design and enrichment programs.

As Kenny (1975) has noted, we can not eliminate all threats to validity with the nonequivalent control group design used in the present research. In Cook and Campbell's (1979, pp. 103-112) discussion of threats to validity in this design, they state that an outcome-four situation exists when the control group scores higher than the experimental group, and that this difference is greater at the pretest than at the posttest (see Fig. 1). The greatest threat to validity here is regression upward. An outcome-four situation occurred in the present experiment for the variables of performance, innovativeness, satisfaction, and integration and boundary-spanning activities. To assess the likelihood of regression, Cook and Campbell suggest an examination of the treatment effect, whether the results are meaningful, and if subjects were assigned to the treatment group because of low pretest scores. As previously noted, it would appear that these outcome-four effects do have rational explanations attributable to the job changes. In addition, pretest scores were not used for the assignment of subjects to the treatment group. Of course, with a nonequivalent control group design one cannot completely rule out regression as an explanation for the outcome-four situations.

Not all the significant results, however, were of the outcome-four type. An outcome-five situation occurs when the experimental group is lower than the control group at the pretest, but crosses over and is higher than the control group at the posttest (see Fig. 1). Cook and Campbell (1979) see this outcome as the most interpretable and the least susceptible to validity threats. Regression is not a very likely alternative explanation for this outcome. The results for role ambiguity and task autonomy are outcome-five situations, while the results for variety approach an outcome-five situation. In addition, when performance was analyzed separately for promotion and nonpromotion subgroups, it was found that the promotion subgroup was higher than the control group at pretest and had a significantly greater increase than the nonequivalent control group, while the nonpromotion subgroup was lower than the control group at pretest and also had a significantly greater increase than the nonequivalent control group. These results would tend to rule out a general regression explanation.



(Based on Cook & Campbell, 1979, pp. 103-112.)

Fig. 1. Generally interpretable outcomes from a nonequivalent control group design.

The results of the present research indicate that a job change for research and development employees can lead to beneficial effects on a number of important variables. These effects appear to hold true for both promotional and nonpromotional job changes. In addition, the results suggest that perceptual measures of job characteristics and role requirements can measure the effects of actual job changes in an organization. Continued work on these instruments seems justified to improve construct validity and to further reduce method variance. Additional support for the present results from other field experiments would also help to substantiate the use of these instruments in the diagnosis and evaluation of job design and enrichment programs.

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